

Project Proposal

Uber and Lyft Comparative Analysis: A Data-Driven Method

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Introduction

With its simple, quick, and affordable transportation options, ride-sharing services are revolutionizing urban mobility while serving to reduce traffic and pollution. Alonso-Mora et al. (2017) developed a mathematical model that illustrates ways to serve 98% of demand with an average wait and trip delay of only 2.8 and 3.5 minutes by optimizing routes based on real-time demand and car locations. Moreover, dynamic pricing techniques are employed by companies such as Uber and Lyft to control supply and demand, which has a big impact on consumer behavior and competitiveness in the market. It is critical to fully understand these pricing strategies since they affect both consumer loyalty and company revenue, especially during times of high demand.

This analysis is vital considering the increasing dependency on ride-sharing services, especially in urban areas. Even slight fluctuations in prices may have a big impact on user behavior and company earnings as millions of rides are taken daily. Customer retention might be impacted, for instance, if surge pricing occurs during days of substantial demand and discourages customers or pushes them to rivals. Since the acceptance of these transportation systems relies on passenger loyalty, studying these dynamics offers helpful data to both customers and service providers and ultimately promotes a more user-friendly and competitive marketplace.

Study objectives

The primary objective of this venture is to evaluate American user preferences and offer data-driven insights to enhance ride-sharing apps such as Uber and Lyft. In addition, the study aims to gain an understanding of what factors have an impact on ride-sharing demand. Through achieving these objectives, the project aims to promote data-driven decisions that can boost operational performance and accessibility of ride-sharing services, eventually enhancing this form of urban transportation for tourists as well as locals.

Expected contributions

It is expected that the knowledge acquired from this study will help develop an understanding of the public transportation picture in the United States, particularly that of ride-sharing services such as Uber and Lyft. This knowledge may be replicated in Canadian landscapes, assist ride-sharing providers in allocating resources, and adapt services to align with user needs to enhance customer satisfaction. Furthermore, by identifying neglected locations, the research may help ensure that ride-sharing services are dispersed evenly. The recommendations might also alert local officials about the necessity for additional amenities or accessibility in these locations. Additionally, the project might draw focus to the shifts in commute patterns that could influence the demand for public transport in the future.

Research questions

The objective of this data analytics project is to thoroughly investigate the datasets from Uber and Lyft to discover additional information regarding ride-sharing dynamics, pricing policies, and customer behavior. To be more specific, the project looks to delve into questions:

1. To determine if users switch between Uber and Lyft based on price differences for similar rides. How does this affect the overall demand for each service?
2. What are the most common distance ranges for rides in the area, and how do these distances correlate with the average fare and cab type (Uber vs. Lyft)?
3. Which ride service, Uber or Lyft, experiences higher demand during peak hours in a specific location?
4. Explore how various factors shape user preferences and popular routes
5. Does day of week/ holidays affect the demand for Uber versus Lyft in a specific location?

References

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